SUMEDH SINHA

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Education

Rutgers, The State University of New Jersey at New Brunswick September 2022 – May 2026
Bachelor of Science (B.S.) in Computer Science, Cum Laude | Dean's List Scholar (2023-2025). GPA: 3.6/4.0
Relevant Coursework: Artificial Intelligence, Operating Systems, Databases, Networks, Data Structures, Algorithms.
Leadership Roles: Technology Chair at Rutgers USACS (2023-2024), Ambassador at SAS Admitted Students Day (2024).

Experience

Verizon x Rutgers University – Wireless Information Network Laboratory October 2025 – Present
Research Assistant | Joint Project North Brunswick, NJ

- Developing **Agentic AI** data analysis workflows to facilitate competitive spatial network management.

Rutgers University – Department of Computer Science August 2023 – May 2026
Teaching Assistant | Introductory CS courses (Top 1% Professor-Nominated TA) New Brunswick, NJ

- Led weekly Python workshops for **50+** students covering programming fundamentals and debugging techniques.
- Evaluated **100+** programming assessments with a 3-day turnaround while ensuring grading consistency and academic integrity.

EssilorLuxottica May 2025 – July 2025
Information Systems Management Intern | Digital Rx Solutions Dallas, TX

- Built a CI/CD pipeline that reduced new software deployment time by **25%** across **45+** North American manufacturing sites.
- Migrated **3** project tracking workflows to Azure DevOps and automated KPI reporting dashboards with Microsoft Power Automate.
- Authored **140+** Python-based regression test cases across **3** test suites to improve release validation coverage.

CGI September 2024 – December 2024
Research Extern | via Rutgers MBS Externship Exchange Program New York, NY (Remote)

- Created a low-cost AI agent triage pipeline using **Ollama** to prioritize and route **1,000+** customer communications.
- Architected a **Flask** dashboard to visualize sentiment and severity scores across **10** classification parameters.

HaltDos July 2023 – August 2023
Software Developer Intern | New Project Ventures Noida, India

- Designed **2** Android apps using Java, enabling the organization to explore mobile development for new product opportunities.

Projects

SnapDance | Learn While You Dance | **$2000 2x Winner at Yale YHack 2026** devpost.com/software/snapdance
Lens Studio, SnapML, JavaScript/TypeScript

- Launched a multiplayer game in Snapchat using **JavaScript** and **SnapML**, taking it from concept to a published app in **24** hours.
- Engineered a real-time scoring engine tracking **18** points, dynamically weighting and normalizing moves to boost accuracy by **35**%.
- Optimized raw ML data to adhere to strict limits of the platform (**100 KB**), while driving user engagement with **3+** aura effects.

CureBytes | Intelligent MedNotes | **$750 Winner at Rutgers HealthHack 2024** github.com/sumedhsinha/health-hack-2024
AWS, JavaScript, HTML/CSS, Google Gemini

- Deployed a web portal with **AWS Amplify** and **JavaScript** to support automated performance tracking for **1100+** students.
- Fine tuned a **Google Gemini** model with **4** custom rubrics for real-time autograding and monitoring by instructional staff.

DocPlus | Document Signing Application | **Senior Year Class Project** github.com/sumedhsinha/DocPlus
Python, MySQL, Docker, Redis, Tkinter

- Reduced read latency by **86%** by engineering a predictive look-ahead caching algorithm using **Python** and **Redis**.
- Processed ∼**2,100 records/sec** by architecting a denormalized **ETL** pipeline with automated quality validation in **MySQL**.
- Programmed a **Tkinter** dashboard to monitor **5+** real-time system parameters like cache hit/miss ratios and end-to-end latencies.

Skills

Languages: Python, Java, C, SQL, JavaScript, TypeScript, HTML/CSS, Bash.
Frameworks/Libraries: React, Next.js, tailwindCSS, OpenCV, Tkinter, FastMCP.
Cloud/Databases: AWS, Supabase, Redis, MySQL Workbench.
Developer Tools: Ollama, ElevenLabs, Docker, Azure DevOps, Android Studio, Salesforce, JIRA, Git VCS, SSH, GDB, Lens Studio.
Hobbies: Aviation Photography/Plane Spotting, Cricket, Traveling, Action Movies.

Volunteer & Community Work

JerseySTEM June 2026 – Present
Software Developer Intern | Grassroots 501(c) nonprofit org bridging the gap in STEM education New Brunswick, NJ (Remote)

- Restructured an oversized **Salesforce** database Object schema through normalization analysis and authored step-by-step documentation for adoption by 200+ volunteers.